Brief explanation of the program

The Fight System is a C++ program that allows the user to get a list of fighters, search for a fighter by ID, and select fighters to battle. Each fighter has attributes such as name, health value, and attack power.During a battle, the fighters take turns dealing randomized damage until one fighter's health reaches zero. Fighter info is loaded from and saved to a text file.

A list of concepts used

- Data types and sizeof()
- Text files
- Pointers
- Arrays
- Binary search
- Strings
- File input/output
- Classes

Screenshots of game running

```
===== FIGHT SYSTEM =====
1. Display all fighters
2. Search for fighter
3. Start a fight
4. Show data type sizes
5. Save and exit
Choice: 1

--- Fighter Roster ---
ID: 101 | Name: Ryu | Health: 100 | Attack: 18
ID: 102 | Name: Ken | Health: 100 | Attack: 17
ID: 103 | Name: ChunLi | Health: 100 | Attack: 16
ID: 104 | Name: Guile | Health: 100 | Attack: 19
ID: 105 | Name: Akuma | Health: 100 | Attack: 22
ID: 106 | Name: Cammy | Health: 100 | Attack: 15
```

```
--- Fighter Roster ---
ID: 101 | Name: Ryu | Health: 100 | Attack: 18
ID: 102 | Name: Ken | Health: 100 | Attack: 17
ID: 103 | Name: ChunLi | Health: 100 | Attack: 16
ID: 104 | Name: Guile | Health: 100 | Attack: 19
ID: 105 | Name: Akuma | Health: 100 | Attack: 22
ID: 106 | Name: Cammy | Health: 100 | Attack: 15

===== FIGHT SYSTEM =====
1. Display all fighters
2. Search for fighter
3. Start a fight
4. Show data type sizes
5. Save and exit
Choice: 2
Enter fighter ID: 105

Fighter found:
ID: 105 | Name: Akuma | Health: 100 | Attack: 22
```

```
===== FIGHT SYSTEM =====
1. Display all fighters
2. Search for fighter
3. Start a fight
4. Show data type sizes
5. Save and exit
Choice: 3
Enter first fighter ID: 101
Enter second fighter ID: 105

=== Ryu VS Akuma ===
Ryu hits Akuma for 8 damage. Akuma HP: 92
Akuma hits Ryu for 15 damage. Ryu HP: 85
Ryu hits Akuma for 16 damage. Akuma HP: 76
Akuma hits Ryu for 16 damage. Ryu HP: 69
Ryu hits Akuma for 9 damage. Akuma HP: 67
Akuma hits Ryu for 19 damage. Ryu HP: 50
Ryu hits Akuma for 8 damage. Akuma HP: 59
Akuma hits Ryu for 16 damage. Ryu HP: 34
Ryu hits Akuma for 17 damage. Akuma HP: 42
Akuma hits Ryu for 8 damage. Ryu HP: 26
Ryu hits Akuma for 14 damage. Akuma HP: 28
Akuma hits Ryu for 9 damage. Ryu HP: 17
Ryu hits Akuma for 2 damage. Akuma HP: 26
Akuma hits Ryu for 20 damage. Ryu HP: 0

Winner: Akuma!
```

```
===== FIGHT SYSTEM =====
1. Display all fighters
2. Search for fighter
3. Start a fight
4. Show data type sizes
5. Save and exit
Choice: 4

--- Data Type Sizes ---
int: 4 bytes
double: 8 bytes
char: 1 byte
Fighter: 36 bytes
```